ICT (information and communications technology, or technologies)

ICT, or information and communications technology (or technologies), is the infrastructure and components that enable modern computing.

Although there is no single, universal definition of ICT, the term is generally accepted to mean all devices, networking components, applications and systems that combined allow people and organizations (i.e., businesses, nonprofit agencies, governments and criminal enterprises) to interact in the digital world.

IT job roles and responsibilities explained

The IT industry is well known for its wide range of job titles and aggrandisement of roles, which can make it hard to pin down exactly what people do. We've decoded some of the more popular positions below in our quick guide to IT roles:

Software engineer

Also known as: application programmer, software architect, system programmer/engineer.

This job in brief: The work of a software engineer typically includes designing and programming system-level software: operating systems, database systems, embedded systems and so on. They understand how both software and hardware function. The work can involve talking to clients and colleagues to assess and define what solution or system is needed, which means there's a lot of interaction as well as full-on technical work. Software engineers are often found in electronics and telecommunications companies. A computing, software engineering or related higher degree is often needed.

Key skills required: Analysis, logical thinking, teamwork and attention to detail.

Systems analyst

Also known as: Product specialist, systems engineer, solutions specialist, technical designer.

This job in brief: Systems analysts investigate and analyse business problems and then design information systems that provide a feasible solution, typically in response to requests from their business or a customer. They gather requirements and identify the costs and the time needed to implement the project. The job needs a mix of business and

technical knowledge, and a good understanding of people. It's a role for analyst programmers to move into and typically requires a few years' experience from graduation.

Key skills include: Ability to extract and analyse information, good communication, persuasion and sensitivity.

Business analyst

Also known as: Business architect, enterprise-wide information specialist.

This job in brief: Business analysts are true midfielders, equally happy talking with technology people, business managers and end users. They identify opportunities for improvement to processes and business operations using information technology. The role is project based and begins with analysing a customer's needs, gathering and documenting requirements and creating a project plan to design the resulting technology solution. Business analysts need technology understanding, but don't necessarily need a technical degree.

Key skills required: Communication, presentation, facilitation, project management and problem solving.

Technical support

Also known as: Helpdesk support, operations analyst, problem manager.

This job in brief: These are the professional troubleshooters of the IT world. Many technical support specialists work for hardware manufacturers and suppliers solving the problems of business customers or consumers, but many work for end-user companies supporting, monitoring and maintaining workplace technology and responding to users'

requests for help. Some lines of support require professionals with specific experience and knowledge, but tech support can also be a good way into the industry for graduates.
Key skills required: Wide ranging tech knowledge, problem solving, communication/listening, patience and diplomacy.

Network engineer

Also known as: Hardware engineer, network designer.
This job in brief: Network engineering is one of the more technically demanding IT jobs. Broadly speaking the role involves setting up, administering, maintaining and upgrading communication systems, local area networks and wide area networks for an organisation. Network engineers are also responsible for security, data storage and disaster recovery strategies. It is a highly technical role and you'll gather a hoard of specialist technical certifications as you progress. A telecoms or computer science-related degree is needed.
Key skills include: Specialist network knowledge, communication, planning, analysis and problem solving.

Technical consultant

Also known as: IT consultant, application specialist, enterprise-wide information specialist.
This job in brief: The term 'consultant' can be a tagline for many IT jobs, but typically technical consultants provide technical expertise to, and develop and implement IT systems for, external clients. They can be involved at any or all stages of the project

lifecycle: pitching for a contract; refining a specification with the client team; designing the system; managing part or all of the project; after sales support... or even developing the code. A technical degree is preferred, but not always necessary.

Key skills include: Communication, presentation, technical and business understanding, project management and teamwork.

Technical sales

Also known as: Sales manager, account manager, sales executive.

This job in brief: Technical sales may be one of the least hands-on technical roles, but it still requires an understanding of how IT is used in business. You may sell hardware, or extol the business benefits of whole systems or services. Day to day, the job could involve phone calls, meetings, conferences and drafting proposals. There will be targets to meet and commission when you reach them. A technology degree isn't necessarily essential, but you will need to have a thorough technical understanding of the product you sell.

Key skills required: Product knowledge, persuasion, interpersonal skills, drive, mobility and business awareness.

Project manager

Also known as: Product planner, project leader, master scheduler.

This job in brief: Project managers organise people, time and resources to make sure information technology projects meet stated requirements and are completed on time and on budget. They may manage a whole project from start to finish or manage part of a larger 'programme'. It isn't an entry-level role: project managers have to be pretty clued up. This requires experience and a good foundation of technology and soft skills, which are essential for working with tech development teams and higher-level business managers.

Key skills required: Organisation, problem solving, communication, clear thinking, and the ability to stay calm under pressure.

Web developer

Also known as: Web designer, web producer, multimedia architect, internet engineer.

This job in brief: Web development is a broad term and covers everything to do with building websites and all the infrastructure that sits behind them. The job is still viewed as the trendy side of IT years after it first emerged. These days web development is pretty technical and involves some hardcore programming as well as the more creative side of designing the user interfaces of new websites. The role can be found in organisations large and small.

Key skills required: Basic understanding of web technologies (client side, server side and databases), analytical thinking, problem solving and creativity

Software tester

Also known as: Test analyst, software quality assurance tester.

This job in brief: Bugs can have a massive impact on the productivity and reputation of an IT firm. Testers try to anticipate all the ways an application or system might be used and how it could fail. They don't necessarily program but they do need a good understanding of code. Testers prepare test scripts and macros, and analyse results, which are fed back to the project leader so that fixes can be made. Testers can also be involved at the early stages of projects in order to anticipate pitfalls before work begins. You can potentially get to a high level as a tester.

Key skills required: Attention to detail, creativity, organisation, analytical and investigative thinking, and communication.

INFORMATION COMMUNICATION TECHNOLOGY (ICT) MANAGER

a) Duties and responsibilities

planning, development, deployment and provision of Information communication and technology services in support of the corporation business. The key tasks will include and not limited to:

i. Formulating and leading in the implementation of the ICT strategy in line with policies, processes & procedures to deliver business objective.

ii. Working with business to define business and ICT performance metrics, and measures, tracks and reports on benefits realization.

iii. Anticipates business needs and identifies and proposes ICT products and services to fulfil those needs

iv. Leading the demand side of the ICT governance on behalf of the corporation and communicates high level business requirements.

v. Planning and leading the corporation in the ICT plans and creating awareness in the functional department on the development of specific systems solutions to meet the functional requirements and corporations objectives

vi. Planning, managing and providing project management (PM) leadership and support for ICT projects using ICT PM methodology.

vii. Managing all ICT systems and ensure that ICT policies are consistently applied and plans for the organization (business recovery plans) BRP and pandemics.

viii. Managing overall systems control by safeguarding the systems against deliberate efforts to fraudulently abusing it through physical and logical systems security and establishment of security features within the system for control purpose, though network monitoring and management

ix. Leading the corporation in the ICT services management, reviews and ensures that agreed ICT services are provided throughout the corporation

x. preparing, managing and monitoring the departments budgets

xi. Leading, motivating and developing the departmental staff to achieve business and people objectives

xii. Managing the department's communication

xiii. Identifying, implementing and benchmarking best practices

xiv. Managing and implementing change initiatives to achieve desired business results

xv. Reviewing ICT Risk management and Business continuity plans b) Requirements for the Job For appointment to this grade an officer must have at least:

i. Master's Degree in any of the following disciplines :- Information Technology, Computer Science/Engineering or Business Information Technology

ii. Bachelor's Degree in any of the following: Information Technology, Computer Science or Business Information Technology from a recognized Institution.

iii. A minimum of ten (10) years experience in relevant work and at least three (3) years in a supervisory role in the Public Service or in the Private Sector;

iv. Professional certification such as Certified Computer Network Professional (CCNP), CCNA, MCSE, MCSA, MCSD, Oracle Certified Database administrator OCDBA, Microsoft Certified databases Administrator MCDBA or their equivalent qualification from a recognized institution.

v. Member of an ICT professional body.

vi. Certificate in Leadership Development Course lasting not less than four (4) weeks from a recognized institution.

vii. Fulfilled the requirements of Chapter Six of the Constitution; and

viii. Demonstrated merit and ability as reflected in work performance and results.

Top 10 Jobs in Information Technology

By

Veronica C.

Information technology – often shortened to just IT – is a buzz phrase you've probably heard ad nauseum if you happen to work with IT personnel or went to school for anything related to computers. IT workers are highly specialized in their field, which is probably why they're often just called "IT nerds." They like what they do and understand it inside and out. IT workers are also essential to just about every modern business model. If a company relies on phones and emails, chances are that there's an IT professional behind it all making sure the cogs in the machine function properly.

So what's the real benefit of being the behind-the-scenes technological lifeblood of a company? Excellent pay is your reward, more often than not. As the economy finds its footing and more jobs open up, expect the requirements of IT professionals to be vast as ever. Curious what you might like to do and how much someone will pay you to do it? Consider these 10 leading fields and their median salary ranges.

1. IT consultant

Ranking number 13 on CNN's most recent Best Jobs in America list, the work of being an IT consultant is as vague as it sounds. In this position, your job is to evaluate the systems and do the research that no one else entirely understands. As CNN puts it, everyone from local startups to the Fortune 500 companies need IT consultants to help them figure out the cheapest and fastest ways to run computers better.

Education: A bachelor's degree in computer science definitely helps, like most positions

on this list. CNN also recommends that an IT consultant specialize in a niche category to help focus on the kind of experience he or she gets.

Salary: $96,400, according to CNN.

2. Cloud architect

You've heard of cloud computing of course. Although that storage space existing in the ether can't be touched literally, it still needs to be organized and given an architecture. That's what this job is for.

Education: A bachelor's degree.

Salary: $112,000, making it one of the highest paying IT jobs.

3. Computer forensic investigator

Computer crime detectives – The Best Schools reports that computer forensic investigators search for, identify and evaluate information from computer systems, often for trial evidence.

Education: TBS says that you'll need a degree in computer forensics, information security or cyber security. Certification from a computer examiner board also helps.

Salary: $64,000 according to TBS.

4. Health IT specialist

Health IT is a blossoming field, especially with major changes going on in healthcare due to the Affordable Care Act and the gradual transition to electronic health records. Health IT specialists will mix computer knowledge will record-keeping skills, but specialties in medical coding, billing and cancer registry are also in demand, according to TBS.

Education: While some health IT jobs require only an associate degree and/or certification, supervisory technician positions call for bachelor's and master's degrees.

Salary: $45,000, according to TBS.

5. Mobile application developer

Chances are that you and most people you know have smartphones and/or tablet computers. According to CareerRealism, the use of mobile tech is predicted to exceed personal computers at some time in 2013, so businesses are more heavily relying on IT professionals with experience in this field than ever before. Using basic coding languages, developers will create programs for future iOS and Android devices.

Education: A bachelor's in software engineering, computer science, mobile computer or related fields, according to TBS.

Salary: $90,000 with high growth outlook, reports TBS.

6. Web developer

Web developers are jacks of all trades. They create web pages, web applications and web content, but their skill set requires them to have excellent understanding of what makes a good operating system, what the average surfer finds visually stimulating and how to optimize sites for mobile tech, among numerous other skills. They also need proficiency in Web languages, like HTML and Javascript.

Education: TBS reports that the road to web design can be learned through accredited degree programs, but many web developers are self taught and use their "portfolios" to win positions.

Salary: $90,000, according to TBS.

7. Software engineer

Like video games? Want to design the next Facebook? This is for you. Software engineers are behind all the programs we run on our mobile devices and personal computers – and there is a very wide range of niche fields you can work in.

Education: According to TBS, a bachelor's degree in software engineering or a related field is best.

Salary: $89,000 according to TBS.

8. Information technology vendor manager

Slightly more hands-off compared to some tech positions, vendor managers oversee supply when it comes to software and hardware. This can mean anything from Microsoft's latest word processor to health IT programs for hospitals.

Education: Computer science degrees are helpful, but a deep understanding of business or even an MBA could clinch a job.

Salary: $88,000, reports TBS.

9. Geospatial professionals

Sound confusing? Geographic information systems are complicated, but exciting and getting moreso every day. GIS tech uses geographic data to evaluate and communicate trends and patterns in visually stylish and comprehensive ways, according to CareerRealism.

Education: Certificate programs and degrees both improve job outlook.

Salary: Up to $84,000, according to CareerRealism.

10. Data Modeler

Another position that translates poorly without jargon, these IT professionals create data designs and define relationships between data fields, according to TBS. Since any company's data is vital, it's modeling needs to work perfectly – a more complex task as reliance on computers grows.

Education: Bachelor's degree in computer science, mathematics or IT – plus on-the-job experience, says TBS.

Salary: A hefty $103,000, according to TBS.